2024 MODIFICATION OF THE ALGORITHM FOR DEFINING POLYGONAL GEOMETRY OF AN OBJECT FOR POLYPOINT TRANSFORMATIONS

27 2026 .

DOI https://doi.org/10.15407/csc.202DOI https://doi.org/10.15407/csc.20222.0.033.xxx.xxx

UDKUDK 514.18 514.18

SydorenkoSydorenko u.Vu.V.., PhD (Eng. sc.), assistant professor, National Technical University of Ukraine “Igor Sikorsky Kyiv Polytechnic Institute”, 37, , PhD (Eng. sc.), assistant professor, National Technical University of Ukraine “Igor Sikorsky Kyiv Polytechnic Institute”, 37, ProspProsp.. PeremohyPeremohy, Kyiv, Ukraine, 03056., , Kyiv, Ukraine, 03056., ORCIDORCID: : https://orcid.org/0000-0002-1953-041https://orcid.org/0000-0002-1953-041

E–mail: suliko3@ukr.netE–mail: suliko3@ukr.net

HorodetskyiHorodetskyi M.V M.V.., , PhD PhD student, National Technical University of Ukraine “Igor Sikorsky Kyiv Polytechnic Institute”, 37, student, National Technical University of Ukraine “Igor Sikorsky Kyiv Polytechnic Institute”, 37, Prosp.PeremohyProsp.Peremohy, Kyiv, Ukraine, 03056., , Kyiv, Ukraine, 03056., ORCID:ORCID: https://orcid.org/https://orcid.org/0000-0003-4673-38940000-0003-4673-3894

EE—mailmail: : HYPERLINK "mailto:horodetskyimykola@gmail.com" horodetskyimykolahorodetsk

, -
,

MODIFICATION OF THE ALGORITHM FOR MODIFICATION OF THE ALGORITHM FOR DEFININGDEFINING POLYGONAL GEOMETRY OF AN OBJECT FOR POLY POLYGONAL GEOMETRY OF AN OBJECT FOR POLYPOINT POINT TRANSFORMATIONSTRANSFORMATIONS

.
. ,
, . ,
, .
, .
, , .
. .

: : , , , -
, , , ’ . , , ’
.

KeywordsKeywords:: polycoordinate polycoordinate mappingsmappings, polypoint transformations, polygon, polypoint transformations, polygonal geometryal geometry, transformation basis, transformation object., transformation basis, transformation object.

, .
,
. , ,
.
, . , -
.
, , . ,
. .
, , ,
« », . , . .
,
,
, , « », .
,
.
. ,
. . ,
, . ,
.
, . , . .
, .
, , ,
, . ,
. , , . ,
., . , , .
.
. . [1]
. ' , [1]
.
, , [2]. [2].
, .
, . , , ,
, [3].
. , .
, , , [3].
, ,
.
, [4]. '
, . ,
[4].
,
, -
,
, [4], ' [4],
' [4], [4], -
[4], [4],
[4] . [4] .
,
.
, , .
, , , .

, ,

, ,

.

-

, ,

..

.. , ,

, .

, , .

, , (R

(R22)) ::

LL11, , AA11, B, B11, C, C11. . ’

, , ,

, , : :

XX ,Y,Y — — - ,- , ββii —

— LL1 1 XX ,Y,Y ..

’ ’ A, B, CA, B, C , ,

„ LL..

1 . 1 .

. . 11. . LL1 1

, .

, . .

, 2.

, . , .

. , 2.

. 2. . 2.

,

.

. ’ , , . ,

. , .

, ,

. . ’ , ,

. ,

. , . , „ .

.

, . ’

, , . ,

.

, .

, . , .

. . : : , ,

, , ,

, ’ , . , , .

, ,

, ’ , . , ’ , .

.

.

- , - , , ,

(. 3). (. 3).

. 3. ' , . 3. ' ,

- , - , , ' , ' .

, ,

.

, ( ), , ( ),

. .

, , : ,

, :

( ). (

).

, ,

, . , .

. . .

.

FIFOFIFO LIFOLIFO,

, ..

, ' .

, ' .

.

.

.

. .

.

, ' .

, ' .

3 3.. , , '

, . .

,

. .

(' ( ))

. .

,

' (x(xii; y; yii), (x), (xjj; y; yjj)) .

.

««ABCDEFGHAABCDEFGHA»» (A -HA) (. (A -HA) (. 44). ,

.). , .

. 4 . 4. .

, ,

,,

. .

(.5) (.5).

.

. . 55. .

, , , (
) , ( ).. -

«A«A11BB11CC11DD11EE11FF11GG11HH11AA11»» ((AA11BB11-
H- H11AA11) ,) , A -HA (. 6 A -HA (. 6).).

. . 66. .

, , . , , -
. «A«A11BB11CC11DD11EE11FF11GG11HH11AA11»»,
,
«ABCDEFGHA», «ABCDEFGHA», .
.

. ,
: .
, :

,,
mm((ββii)) – – . , .
, mm - - ββ .
,
, . , ,
' . . , ,
. , , ' .
,
,
, , ' . ' . , 4 '
, 4 ' (.7 (.7).).
. 7. 7. ' . '

7 7, ' . , ' ,
, ,
. ,
, . ,
.
, ,
..
.
, , .
,
, .
, , .
, , , . .

,
. . .
. - .: , 2016. - .92- .118-125. . . '
. . - .: , 2016. - .92- .118-125.
'
. ., . . .
,
. ., . . .
. 2019. - .16.- .17-24. . 2019. - .16.- .17-24.
' ,
. ., . ., ...
. . 2020.-

.17.- .108-114. . . 2020.- .17.- .108-114.

. ., ..., . .

. . 2020, . .
2020, .97 —97 — .129-135. .129-135.

’
. ., . ., . . .
. ., . ., . . ’ . -
. 2022. 102. . .65-75 . 2022. 102. . .65-75

, ., , ., & , . (2023). ’ . ,
., , ., & , . (2023). ’ .
, (25), 202-209 , (25), 202-209

Sidorenko Yu, Sidorenko Yu, KryvdaKryvda O., O., LeshchynskaLeshchynska I. System of modeling of structural elements of ventilation systems by I. System of modeling of structural elements of ventilation systems by polyco rdinatepolyco rdinate transformations. Strength of Materials and Theory of Structures. 2020. No. 104. P. 221-228. transformations. Strength of Materials and Theory of Structures. 2020. No. 104. P. 221-228.

. ., . ., ...
. . 2019. –
.15. – . 151-161. . . 2019. – .15. – . 151-161.

REFERENCESREFERENCES

1.1. SidorenkoSidorenko YuYu..VV. . SistemaSistema modelyuvannyamodelyuvannya geometrichnikhgeometrichnikh obob"yektiyekti``vv zaza dopomogoyudopomogoyu polipoli``tochkovikhtochk
peretvorenperetvoren`.`. PrikladnaPrikladna geometri`yageometri`ya ta ta i`nzhenernai`nzhenerna grafi`kagrafi`ka. - . - K.:KDTUBAK.:KDTUBA, 2016. - Vip.92- s.118-125., 2016. - Vip.92- s.118-125.

2.2. Sidorenko Sidorenko Yu.VYu.V., ., BadayevBadayev Yu.IYu.I`.`. GeometrichneGeometrichne modelyuvannyamodelyuvannya skladnikhskladnikh ob'yekti`vob'yekti`v nana osnoviosnovi` ` poli`tochkovikhpoli`tochkovikh vi`dobrazhenvi`dobrazhen` ` vi`dri`zki`vvi`dri`zki`v pryamikhpryamikh. . SuchasniSuchasni` ` problemiproblemi modelyuvannyamodelyuvannya. 2019. -Vip.16.-S.17-24. . 2019. -Vip.16.-S.17-24.

3.3. Sidorenko Sidorenko Yu.VYu.V., ., ShaldenkoShaldenko O.V. O.V. Vi`zuali`zaczi`yaVi`zuali`zaczi`
ob'yekti`vob'yekti`v poli`tochkovikhpoli`tochkovikh peretvorenperetvoren` za ` za dopomogoyudopomogoyu i`nterpolyaczi`jnoyii`nterpolyaczi`jnoyi funkczi`yifunkczi`yi GausaGausa. . SuchasniSuchasni` ` problemiproblemi modelyuvannyamodelyuvannya. 2020.- Vip.17.-S.108-114. . 2020.- Vip.17.- S.108-114.

4.4. Sidorenko Sidorenko Yu.VYu.V., ., Zalevs`kaZalevs`ka O.V. O.V. Pi`dvishhennyaPi`dvishhennya tochnostitochnosti` ` algoritmualgoritmu poli`tochkovikhpoli`tochkovikh peretvorenperetvoren`. `. PrikladnaPrikladna geometri`yageometri`ya ta ta i`nzhenernai`nzhenerna grafi`kagrafi`ka. 2020, vip.. 2020, vip.97 —97 — S.129-135. S.129-135.

5.5. Sidorenko Sidorenko Yu.VYu.V., ., Zalevs`kaZalevs`ka O.V., O.V., ShaldenkoShaldenko O. V. O. V. ObchislennyaObchislennya ploshhiploshhi` ` peretvorenogoperetvorenogo ob'yektaob'yekta pripri poli`tochkovikhpoli`tochkovikh peretvorennyakhperetvorennyakh. . PrikladnaPrikladna geometri`yageometri`ya ta ta i`nzhenernai`nzhenerna grafi`kagrafi`ka. 2022. . 2022. VipuskVipusk 102. 102. KiyivKiyiv. St.65-75 . St.65-75

6.6. Sidorenko, Yu., Sidorenko, Yu., ShaldenkoShaldenko, O., & , O., & Onis`koOnis`ko, A. (2023). Sistema , A. (2023). Sistema poli`tochkovikhpoli`tochkovikh peretvorenperetvoren` ` grupigrupi ob'yekti`vob'yekti`v v v odnomuodnomu bazisibazisi`.`. SuchasniSuchasni` ` problemiproblemi

modelyuvannyamodelyuvannya, (25), 202-209 , (25), 202-209

7.7. Sidorenko Yu, Sidorenko Yu, KryvdaKryvda O., O., LeshchynskaLeshchynska I. System of modeling of structural elements of ventilation systems by polycoordinate transformations. Strength of Materials and Theory of Structures. 2020. No. 104. P. 221-228. I. System of modeling of structural elements of ventilation systems by polycoordinate transformations. Strength of Materials and Theory of Structures. 2020. No. 104. P. 221-228.

8. 8. Sidorenko Sidorenko Yu.VYu.V., ., ShaldenkoShaldenko O.V. O.V. VagoviVagovi` ` poli`tochkovipoli`tochkovi` ` peretvorennyaperetvorennya pripri modelyuvannimodelyuvanni` ` mozhlivikhmozhlivikh rezul`tati`vrezul`tati`v plastichnoyiplastichnoyi khi`rurgi`yikhi`rurgi`yi. . SuchasniSuchasni` ` problemiproblemi modelyuvannyamodelyuvannya. 2019. – Vip.15. – S. 151-161.. 2019. – Vip.15. – S. 151-161.

. . . . „ . . . . , , , , -
« », « -
», 03056, . , . , 37, , 03056, . , . , 37, -
, ORCIDORCID: : httpshttps://:/orcidorcid..orgorg// HYPERLINK "https://orcid.org/0000-0002-1953-041" 0000-0002-1953-0410000-0002-1953-041 sulikosuliko3@3@ukrukr..netnet

. . . . „ , , -
« », « -
», 03056, . , . , 37, , 03056, . , . , 37, , ORCIDORCID: :
httpshttps://:/orcidorcid..orgorg//0000-0003-4673-38940000-0003-4673-3894 horodetskyimykolahorodetskyi

, -
,

. .
' , ' ,
, , .
, , .
'
,
. , ,
. ' ,
,
, ' . ' .
,
. ,
.

. .
, , , ,
, .
, .
, -
..
,
, ,
,
„ .
, .

. , . , ,
, , ,
,
,

- , . ,
- , .

: : , , , , ’ -
. , , , , , .